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NATIONAL INSTITUTE OF TECHNOLOGY: TIRUCHIRAPPALLI: 620 015
DEPARTMENT OF MECHANICAL ENGINEERING
MEIR11 – Basics of Mechanical Engineering

Branch: Instrumentation and Control Engineering
Date: 14.11.2025 Class Quiz – 1

Semester: I
Time: 1.30 PM -1.45 PM

Maximum Marks:10

1. The type of condenser in window AC
☒ a) Natural circulated air-cooled *there is a fan*
b) Forced circulated air-cooled
c) Water-cooled
d) Liquid cooled.
2. Choose the false statements
i. Two-stroke engines are more powerful compared to 4 strokes of the same capacity *(True)*
ii. Diesel engine maintenance is less compared to petrol engine *(False)*
iii. Modern petrol cars with petrol injection has no carburetor. *(True)*
iv. The time available for the exhaust process is less in the case of four-stroke engines compared to 2 stroke engines. *(False)*
a) iii, iv b) ii, iv ☒ c) i, iv d) i, iii
3. The purpose of a fuel injector in a diesel engine is
a) to meter the quantity of diesel to be supplied
b) to increase the pressure of diesel
c) to atomize the diesel and distribute the diesel into the cylinder
☒ d) all of the above *fuel injection pump*
4. A refrigerator of 7.5 ton capacity operates on the reversed Carnot cycle. The temperature limits are 35°C and -20°C . Calculate the COP of the system and power required respectively.
2.5 x 3.5 kW
☒ a) 4.6, 5.7 kW b) 3.6, 5.7 kW c) 5.7, 4.6 kW d) 4.6, 7.5kW *2250K*
5. A condenser of a refrigeration system rejects heat at a rate of 120kW, while its compressor consumes a power of 30kW. The COP of the refrigeration for this system would be
120kW
a) 2 b) 3 ☒ c) 4 d) 5 *21-22*
6. Match the following
- | | |
|----------------|-------------|
| I. Forging | i. Profiles |
| II. Rolling | ii. Gears |
| III. Extrusion | iii. Plates |
| IV. Drawing | iv. Wires |

a) I-iii, II-iv, III-ii, IV-i

b) I-ii, II-iii, III-i, IV-iv

☒ c) I-i, II-iii, III-ii, IV-iv

d) I-iv, II-ii, III-iii, IV-i

7. Choose a process that squeezes metal into peaks and troughs with plastic deformation.

☒ a) Knurling

b) Turning

c) Facing

d) Grooving

8. In Up-Milling the direction of cutter rotation and table feed are

a) Same Direction

☒ b) Opposite Direction

c) Perpendicular to each other

d) Depending on cutter

9. A compound gear train has gears A-B-C-D where A drives B, B and C are on same shaft, and C drives D. If $A:B=1:3$, and $C:D=1:4$, then the overall speed ratio (A:D) is

a) 1:7

☒ b) 1:12

c) 1:6

d) 1:10

10. In a differential, when a vehicle takes a right turn

a) Both wheels rotate at same speed

☒ b) Left wheel rotates faster than right wheel

c) Right wheel rotates faster than left wheel

d) Speed of both wheels decrease equally